



Fig. 5: EUV lithography

Next-generation transistor is called nanowire FET, which is Fin-FET turned on its side with a gate wrapped around it. Nanowire FET, sometimes called a gate-all-around FET, is said to meet the device requirements for 5nm, as defined by International Technology Roadmap for Semiconductors. It has paved the idea of making switch-in-transistor architectures down the road and is developing technologies in the arena.

Chipmakers see a path to extend today's FinFETs to 7nm, but 5nm is far from certain and it may never happen. Indeed, there are a multitude of technical challenges at 5nm. And the cost for 5nm is expected to be astronomical. Performance and cost concerns are the big challenges in scaling to 5nm, and addressing these will involve extension of current approaches as well as introduction of new technologies and materials. If the industry moves forward with 5nm, and so to help the industry get ahead of the curve, semiconductor engineering has assembled a list of some of the more challenging process steps at 5nm.

Patterning and mask making

For 7nm and beyond, patterning is the biggest challenge. This technology has to be production worthy, with the right uptime of tools and an economic throughput per day. In theory, EUV simplifies the pat-

terned process. With 193nm immersion and multiple patterning, there are 34 lithography steps and 60 metrology steps at 7nm. This compares to just six lithography steps and seven metrology steps for 28nm. With EUV, there are just nine

lithography steps and 12 metrology steps at 7nm. Even so, chipmakers still require both EUV and multiple-patterning at 7nm and beyond. And ultimately, the decision to put EUV into production depends on the maturity of the power source, mask infrastructure and resists.

Today, oft-delayed EUV source can generate 80 watts of power. But chipmakers want 250 watts to bring EUV into mass production. It has to show reliability and availability. There are other issues as well. As lines become narrower at each node, the industry faces a growing and problematic issue called line-edge roughness. Basically, line-edge roughness is a deviation on the edge of a line. It is a line-width variation that does not scale with feature size. Meanwhile, as before, lithography is tied to the photomask.

At 7nm, photomask makers may need to prepare for both traditional optical and EUV masks. Optical masks are complex and expensive at 7nm. Because of the increased demand on the process window for the wafer, shapes drawn on masks are becoming smaller. EUV mask shapes are less complex than ArFi (193nm ArF immersion) at those nodes.

But EUV brings other issues on the mask. For example, correction for shadowing and other effects at wafer level has the potential to

MECO®

SINCE 1962

PROGRAMMABLE DIGITAL PANEL METERS



SNP9635SS



Features

- TRMS Measurement
- Easy Programming
- 4 Digit Super Bright Red LED Display
- High Accuracy
- Setup / Programming Protected by Password
- Dual Aux. 110V AC/230V AC
- 1A / 5A AC Input in the same Meter (User Selectable) for Ammeter
- 50-500V AC Input for Voltmeter
- User Programmable Display from 0-9999 as per the CTP / PTR
- Auto Selection of Decimal Point

An ISO 9001:2008 Certified Company

MECO INSTRUMENTS PRIVATE LTD.

Plot No. EL-1, MIDC Electronic Zone,

TTC Industrial Area, Mahape,

Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)

Fax : 0091-22-27673310, 27673330

Email : sales@meconist.com

Web : www.meconist.com